Title: High-Fidelity Real-Time Robotic Ultrasound Simulation via Semantic-Guided Generative Translation

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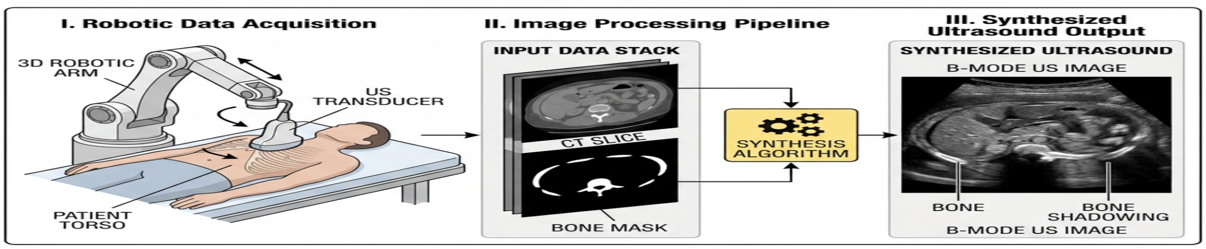

Robotic ultrasound (US) imaging offers a promising solution for standardized, operator-independent diagnostic scanning. However, developing robust control policies requires high-fidelity, computationally efficient simulation environments that can accurately mimic both contact dynamics and realistic US image generation. In this work, we present a lightweight, real-time, and motion-adaptive robotic ultrasound scanning simulation framework built in PyBullet, utilizing a clinical curved array transducer probe model mounted on a Franka Panda manipulator. To bridge the gap between simulation and real B-mode US scans, we propose a two-channel semantic-guided generative translation pipeline that combines oblique CT slices with bone segmentation masks. By incorporating intensity histogram matching and highly optimized linear interpolation pathways, the framework delivers realistic ultrasound synthesis with sharp acoustic shadowing and bone reflections at negligible computational overhead, running entirely on standard CPU hardware. Furthermore, we wrap the simulator in a standard OpenAI Gymnasium environment featuring continuous 6-DOF controls, contact force estimation, and safety-critical terminations, establishing an accessible, high-performance platform for training and validating autonomous robotic scanning policies without the need for local GPU acceleration.